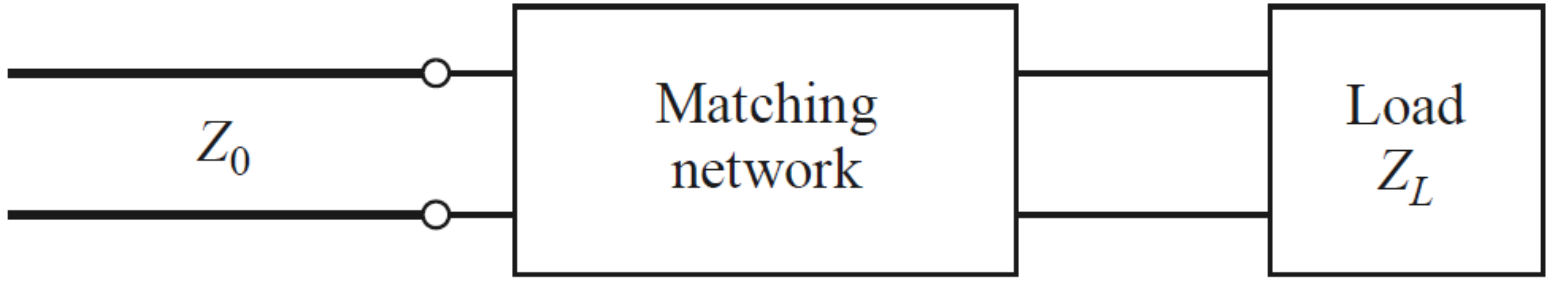
# EEME 30004
# Communications & Networks

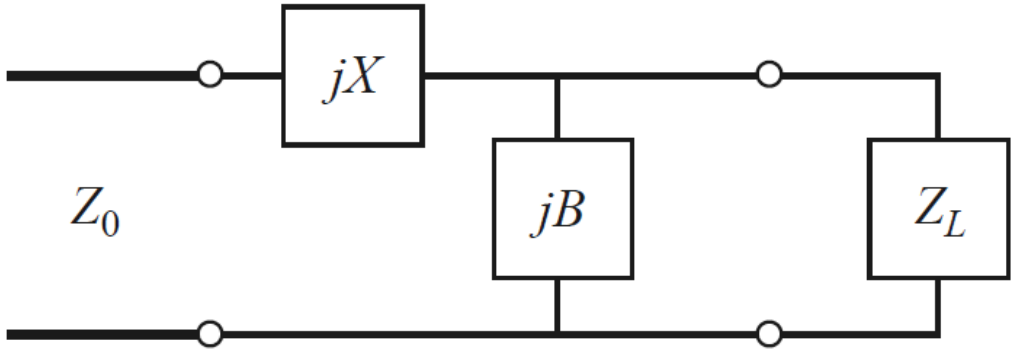
# Impedance matching Part 2

Dr Sean Gu

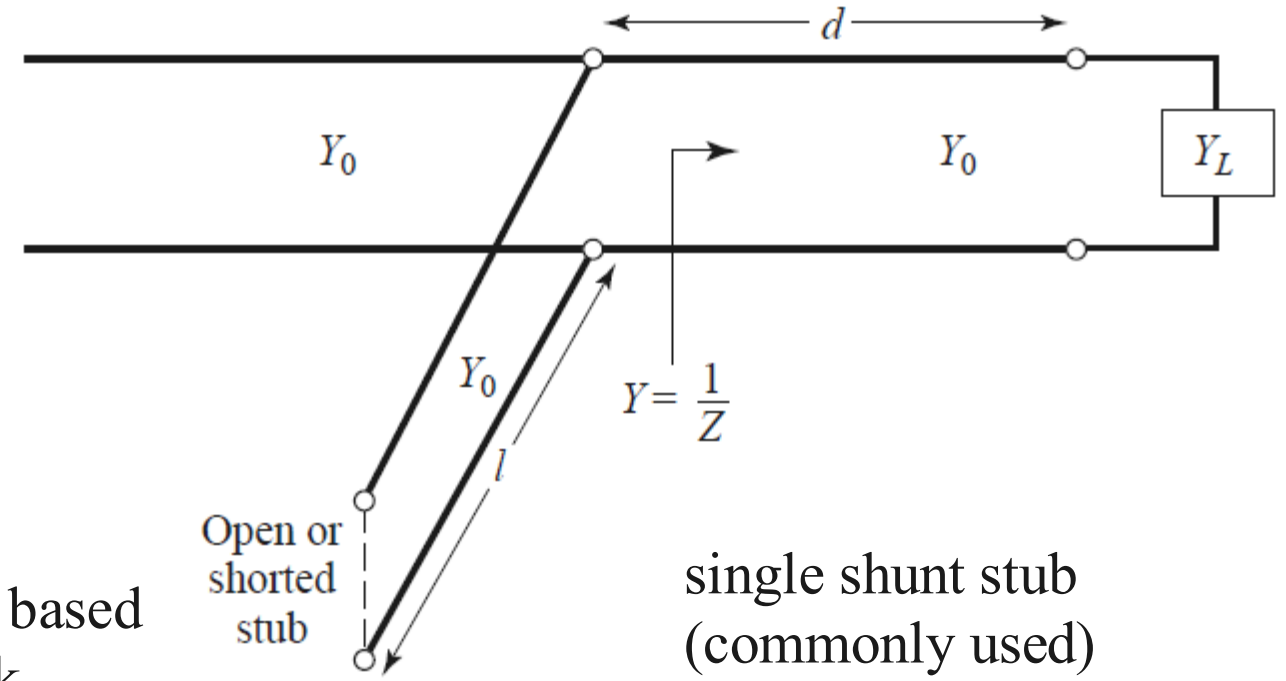
# Impedance matching

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*LC* based matching network

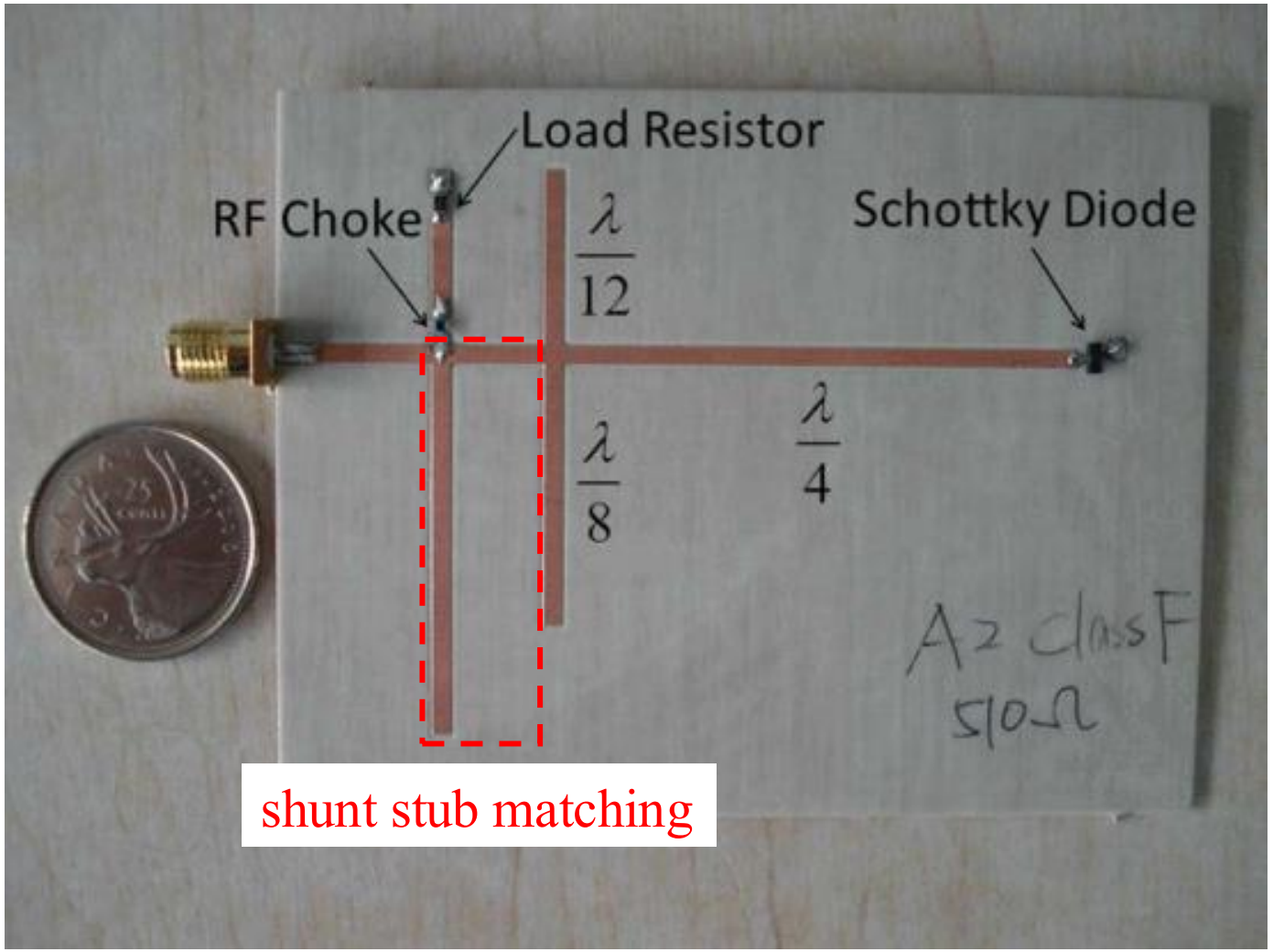


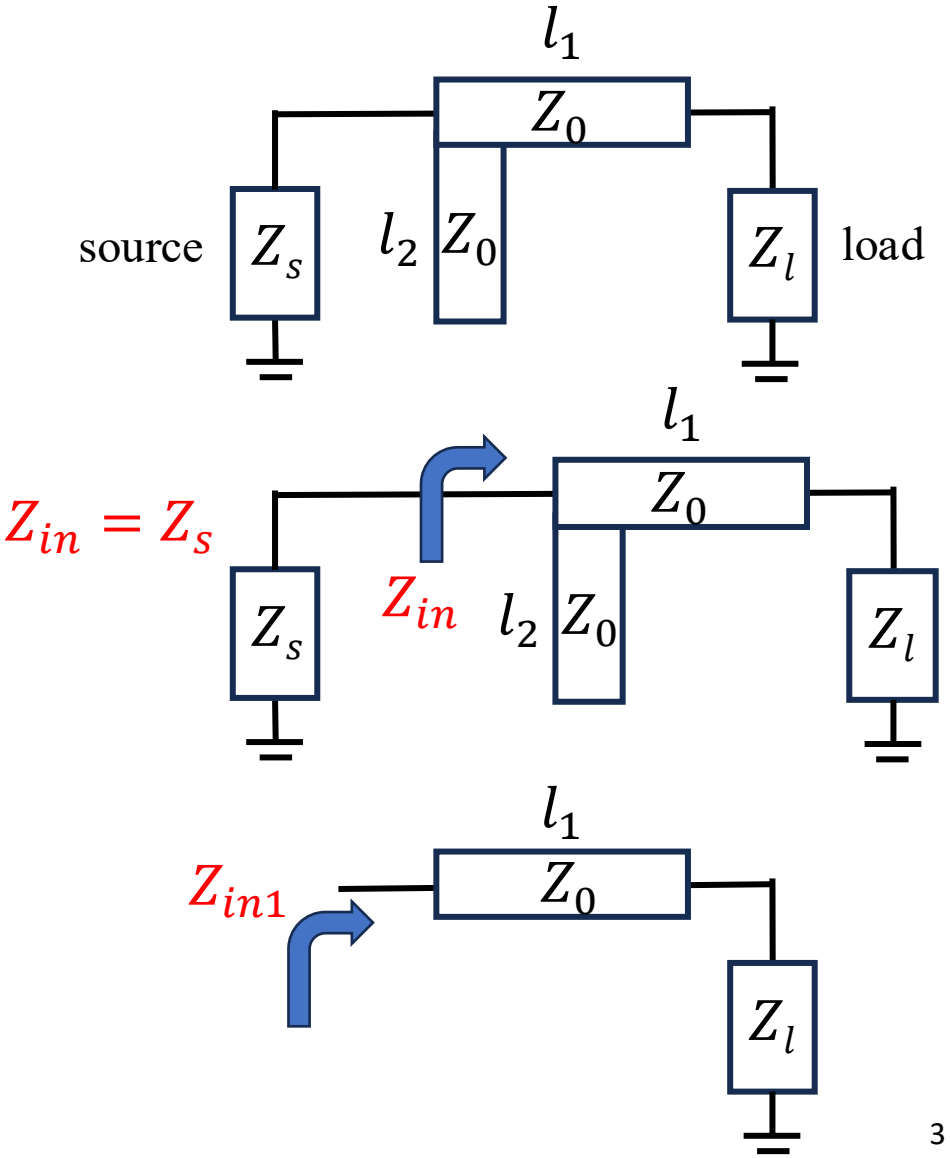

transmission line based matching network

single shunt stub (commonly used)

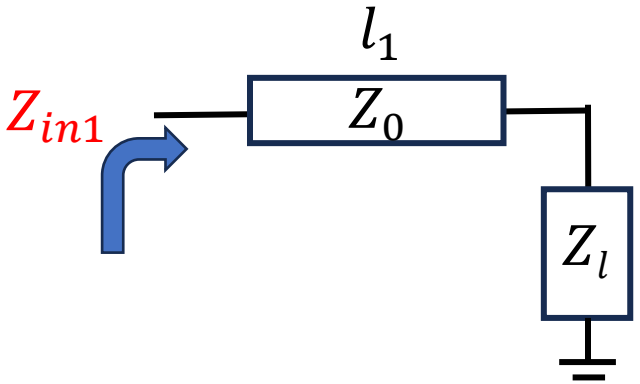
# Impedance matching

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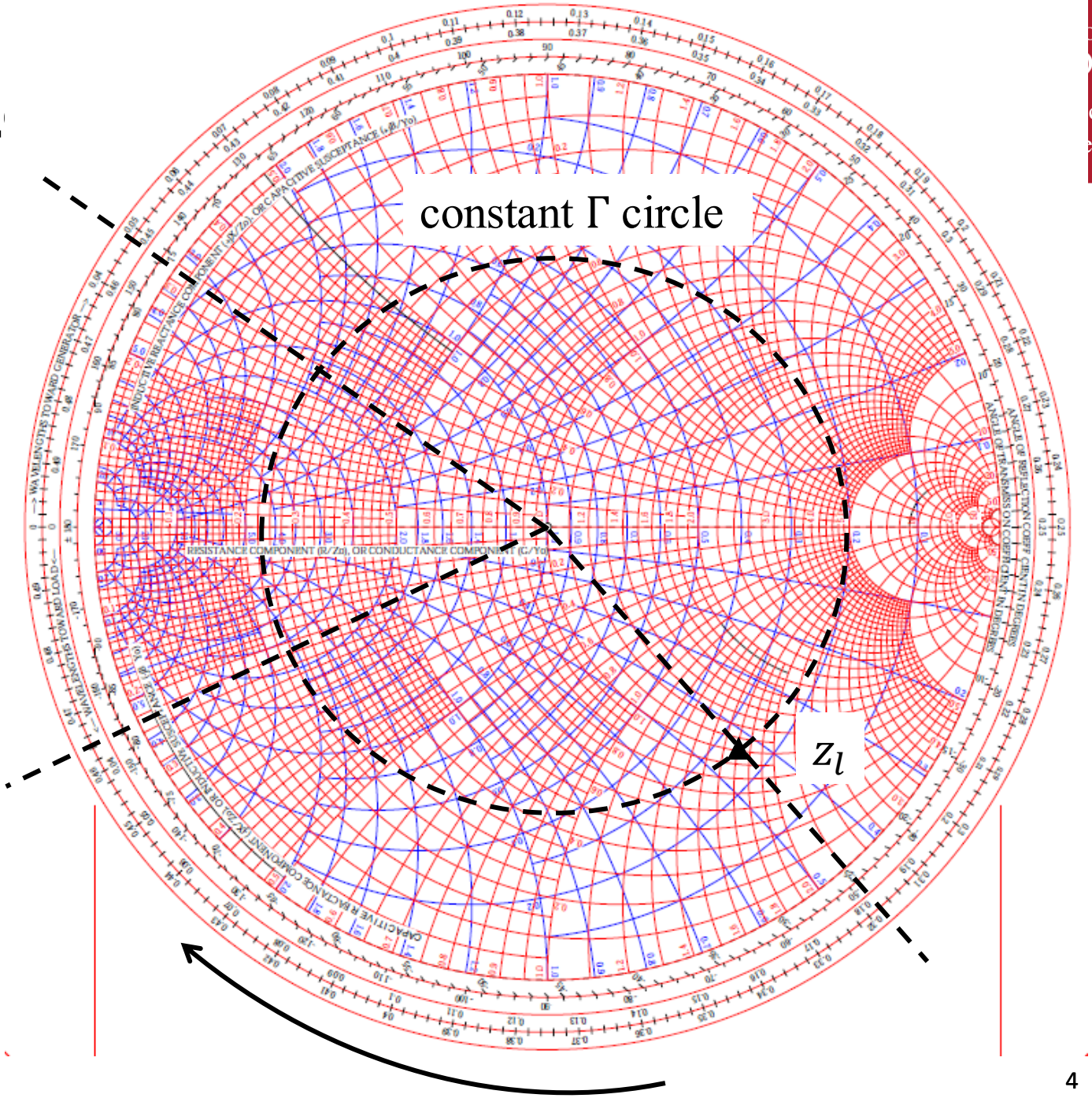



# Impedance matchin

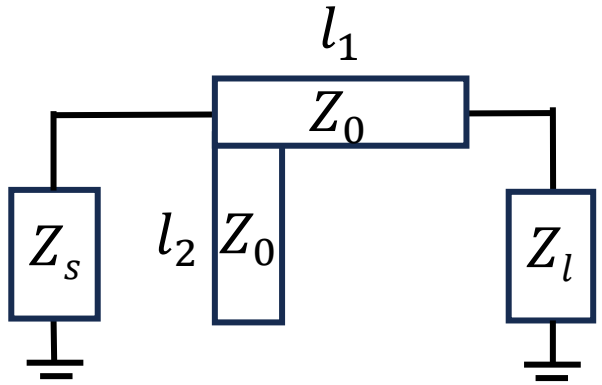


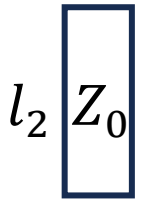



Where do we stop rotating?

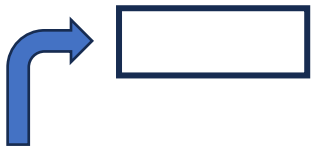


# Impedance matchin




stub can be open-circuited or short circuited



open-circuited

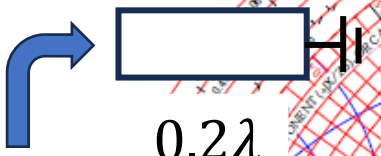


short-circuited

$y_{in} = jX$

$y_{in} = -j0.32$

short-circuited

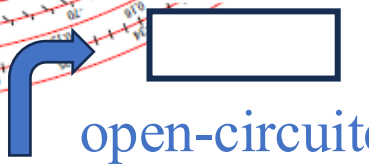


$0.2\lambda$

short

open

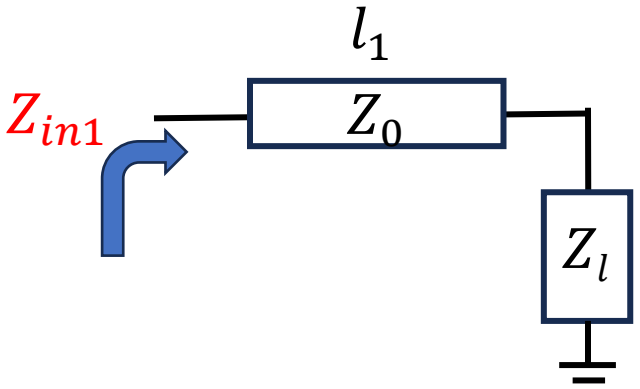
$0.06\lambda$



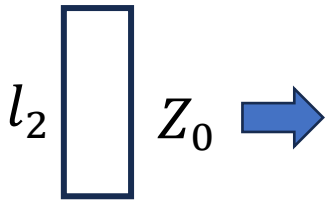
open-circuited

$y_{in} = j0.4$

# Impedance matchin



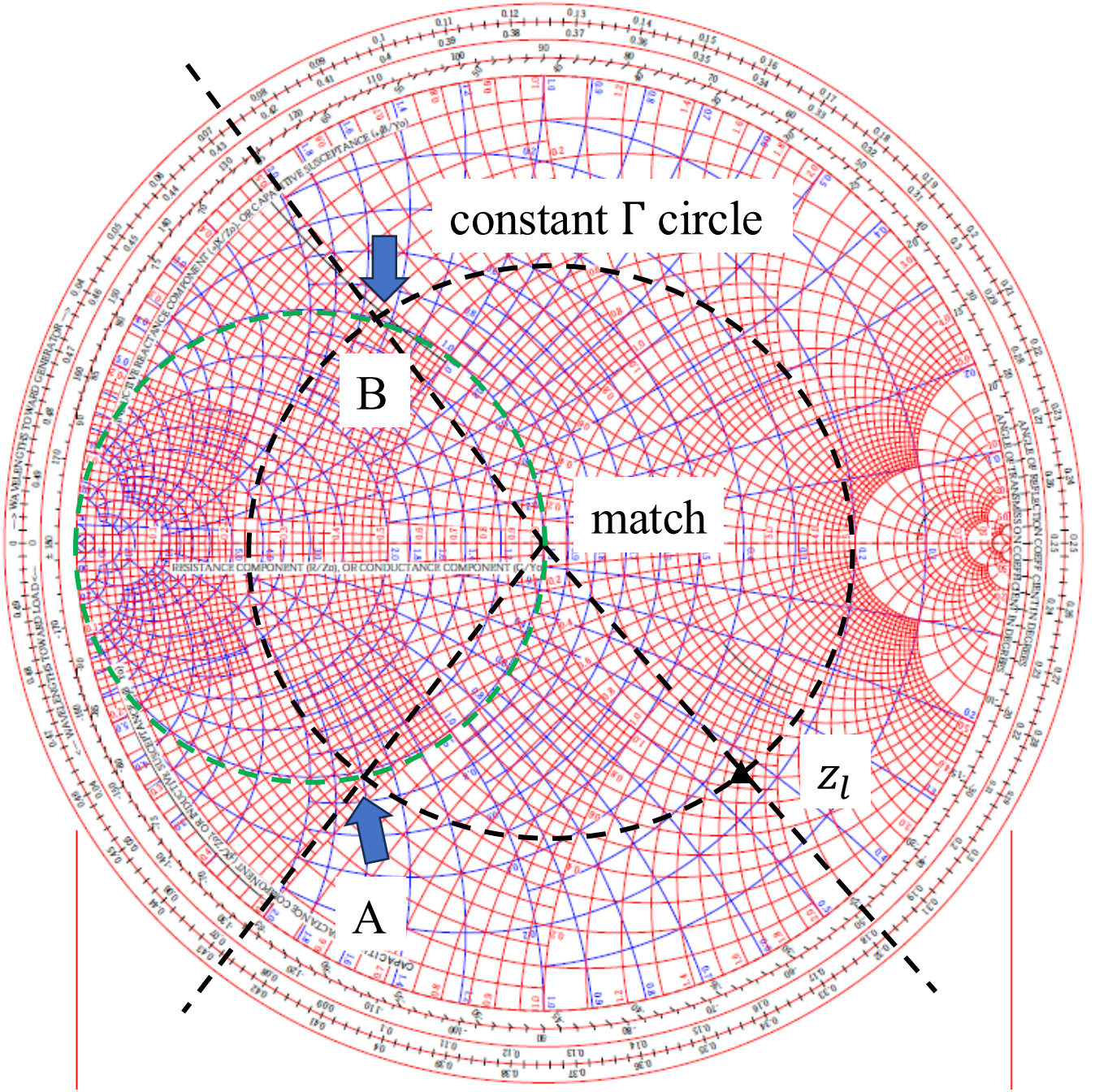

$$y_{in1} = \frac{1}{z_{in1}} = 1 + jX \text{ or } 1 - jX$$



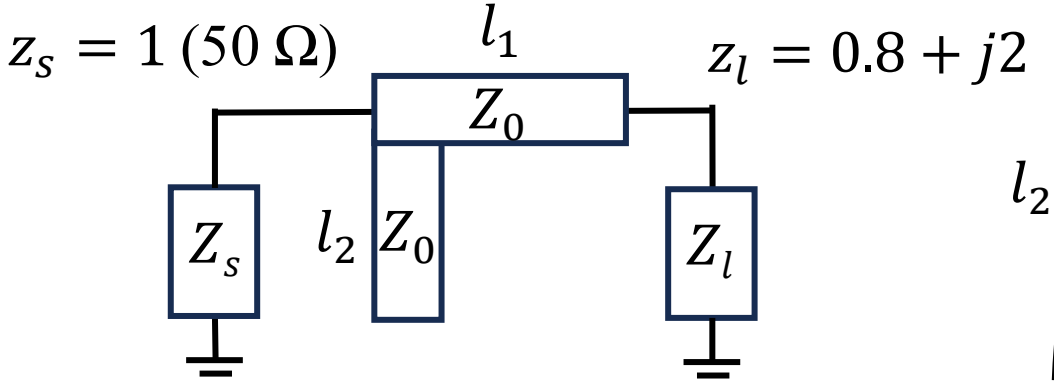

can cancel $\pm jX$

A: $y_{in1} = 1 + jX$

B: $y_{in1} = 1 - jX$



# Impedance matchin



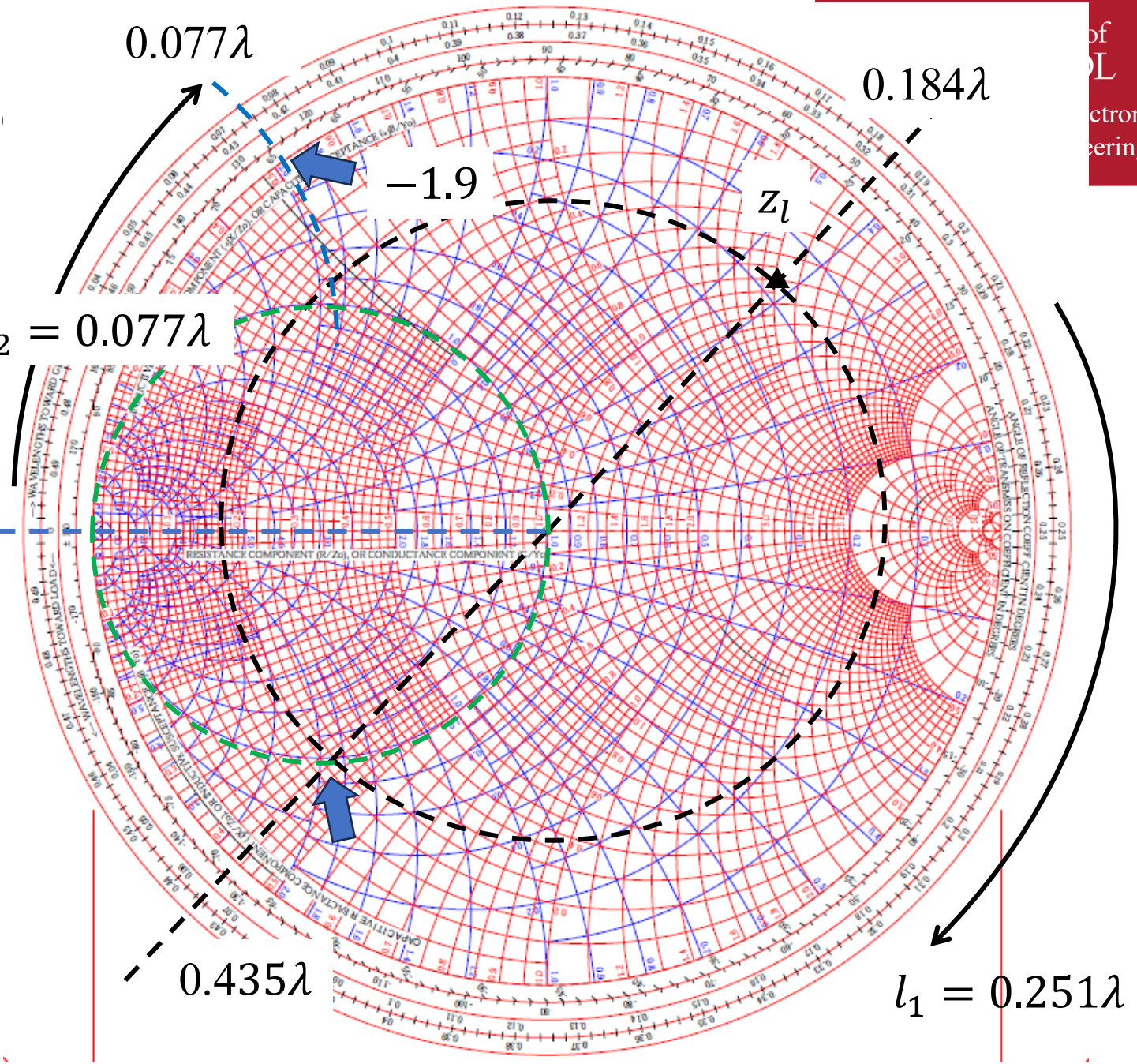

step 1: plot $z_l$, find initial location

step 2: rotate $z_l$ and find the interception point with $y = 1$ circle, take down location info

step 3: calculate $l_1 = 0.435\lambda - 0.184\lambda = 0.251\lambda$

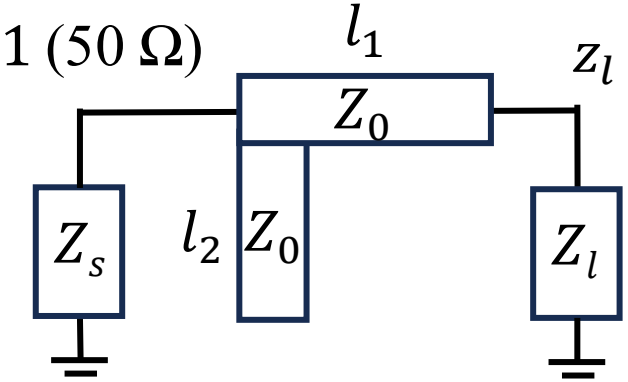
step 4: read $y_{in} = 1 + j1.9$

step 5: create a shunt stub with $y_{stub} = -j1.9$, **short** shunt stub ($0.077\lambda$)

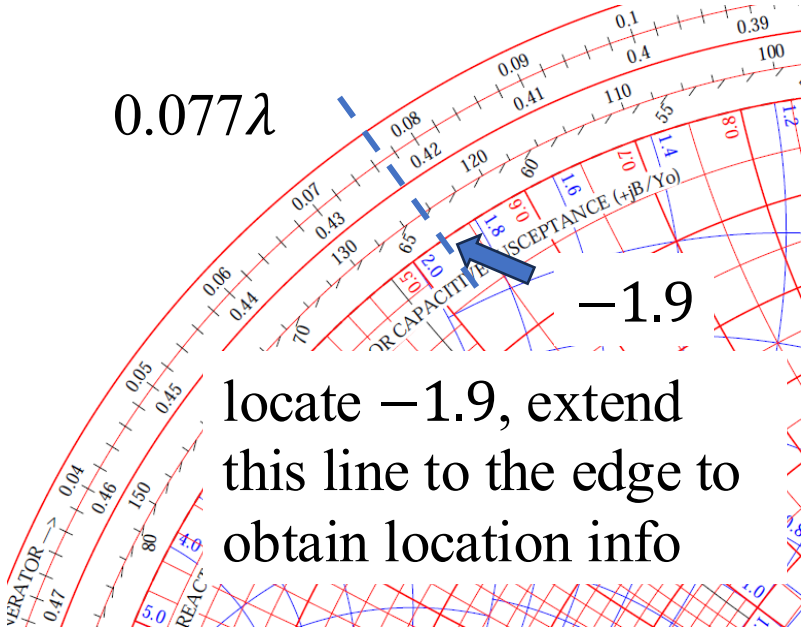


# Impedance matchin

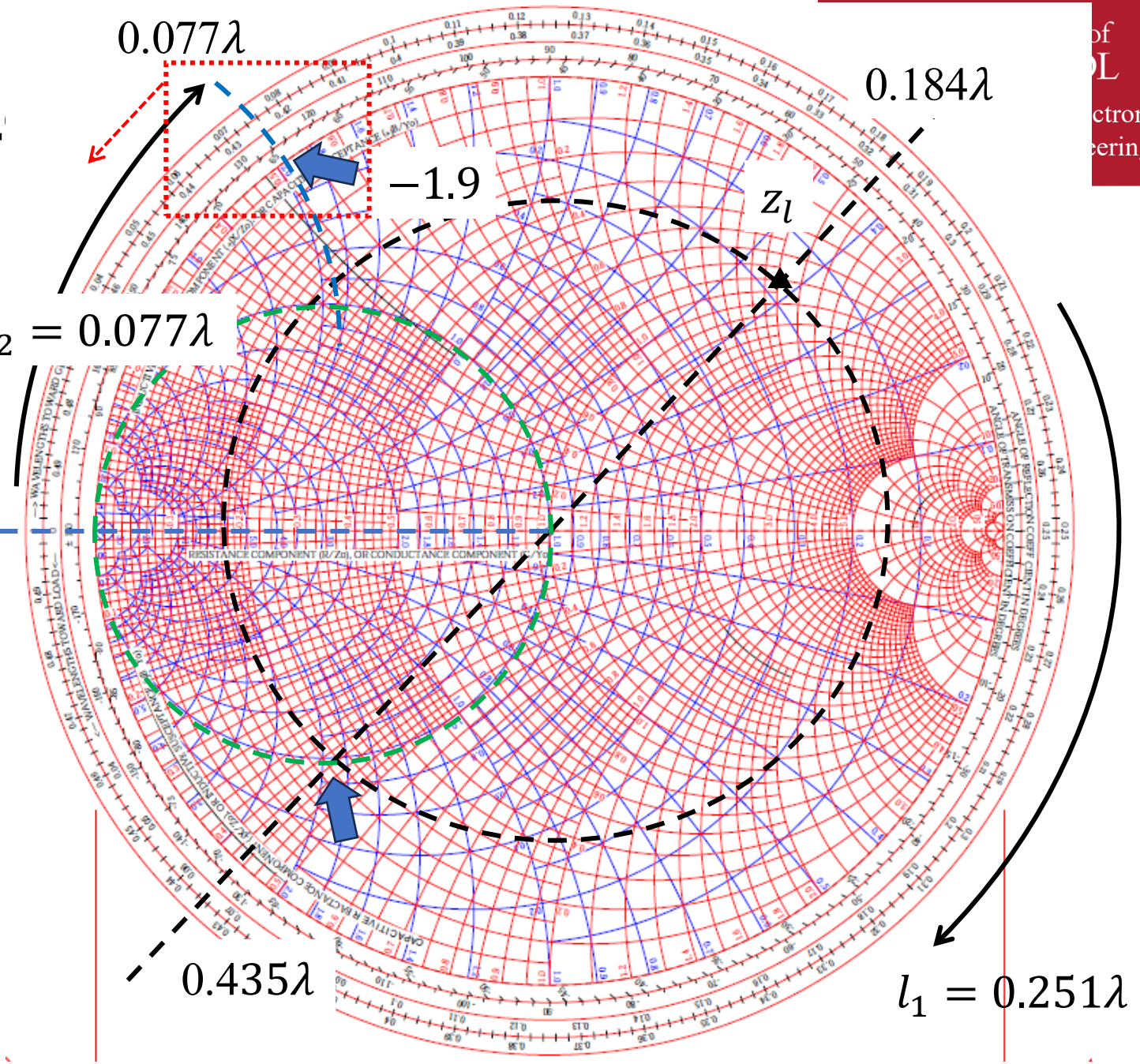
$z_s = 1\ (50\ \Omega)$ $l_1$ $z_l = 0.8 + j2$




step 5: create a shunt stub with $y_{stub} = -j1.9$, short stub $(0.077\lambda)$




locate $-1.9$, extend this line to the edge to obtain location info



# Impedance matching

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Online Smith Chart tool (https://www.will-kelsey.com/smith_chart/)

step 5: create a shunt stub with $y_{stub} = -j1.9$, short stub $(0.077\lambda)$



from electrical length to actual microstrip line length



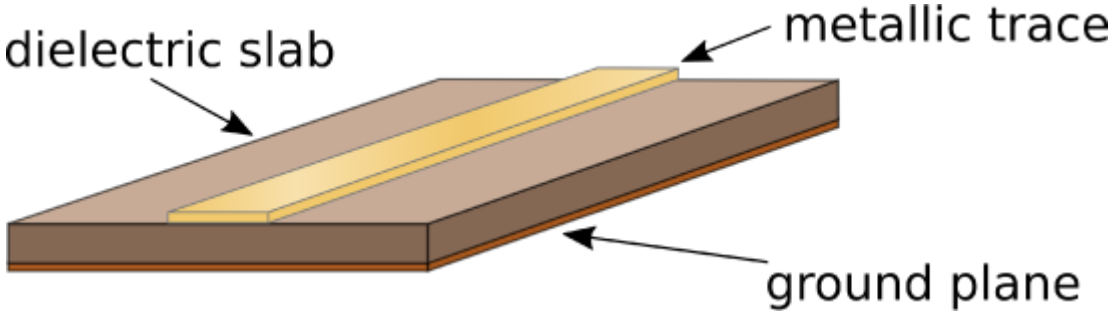
$$\lambda = \frac{c}{f\sqrt{\varepsilon_{eff}}}$$

effective permittivity or dielectric constant



assume the microstrip line is built on Teflon substrate ($\varepsilon_{eff} = 1.8$), operating frequency is 2 GHz, determine the actual shunt stub length

microstrip transmission line




$$\lambda = \frac{c}{f\sqrt{\varepsilon_{eff}}} = \frac{3\times10^8}{2\times10^9\times\sqrt{1.8}} = 0.112 \text{ m or } 112 \text{ mm}$$

$$0.077\lambda = 112\times0.077 = 8.6 \text{ mm}$$

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# Questions?

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